Test Cases and SRS Traceability Update

Hostel Management System

Test-case update required by the Software Engineering Mini-Project brief

Project	Hostel Management System
Document Version	1.0
Prepared for	Software Engineering Mini-Project

1. Purpose and Relationship to the Test Plan

This document provides the detailed test cases referenced by the Test Plan. The cases cover functional and non-functional requirements and include dedicated security validation. Execution status is intentionally recorded as "Not Executed" because these documents define the test baseline rather than claim completed testing.

2. Test Case Conventions

Field	Meaning
Test Case ID	Unique identifier referenced by the Test Plan.
Requirement ID	SRS requirement verified by the test.
Preconditions	State required before execution.
Test data	Synthetic data to use.
Expected result	Observable acceptance condition.
Status	Planned baseline status before execution.

3. Functional Test Cases

ID	Req.	Test case	Preconditions	Test data	Steps	Expected result	Status
TP-01	FR-01	Valid authentication	Registered account exists.	Valid user credentials.	Submit correct identifier/password.	Protected functions become available; an authenticated session is established.	Not Executed
TP-02	FR-02	Create/update student	Administrator is authenticated.	New student identifier and valid details.	Create a record, then update one field.	Record is created; subsequent retrieval shows the updated value.	Not Executed
TP-03	FR-03	Search student	Administrator is authenticated; record exists.	Existing student_id.	Search by student identifier.	Matching student record is displayed; unrelated record is not returned.	Not Executed
TP-04	FR-04	Create/update room	Administrator is authenticated.	Room ID and positive capacity.	Create room, then change capacity.	Room is stored with unique ID and updated capacity.	Not Executed
TP-05	FR-05	Valid room allocation	Administrator authenticated; student has no active allocation; room has capacity.	Student S001, Room R101 with available capacity.	Submit allocation.	Allocation is created and occupancy increases by one.	Not Executed
TP-06	FR-06	Reject full-room allocation	Administrator authenticated; selected room is at capacity.	Full room and unallocated student.	Attempt allocation to full room.	Request is rejected with conflict; no new allocation is stored.	Not Executed
TP-07	FR-06	Reject duplicate student allocation	Administrator authenticated; student has active allocation.	Student with existing active allocation and	Attempt second allocation.	Request is rejected with conflict; existing allocation remains unchanged.	Not Executed

ID	Req.	Test case	Preconditions	Test data	Steps	Expected result	Status
				another room.			
TP-08	FR-07	Modify/deallocate allocation	Administrator authenticated; active allocation exists.	Existing allocation.	Modify room where valid; then deallocate.	Valid modification updates allocation; deallocation removes active status and releases capacity.	Not Executed
TP-09	FR-08	Occupancy display	Authenticated user; room and allocation data exist.	Rooms with known capacities and allocations.	Open room occupancy view.	Displayed occupied/available counts match database state.	Not Executed
TP-10	FR-09, SEC-02	Student self-view and isolation	Student account A and student account B exist.	Two student records with different allocations.	Login as A and access own and B's resource.	A can view own allowed data; B's protected resource is denied.	Not Executed

4. Non-Functional and Security Test Cases

ID	Req.	Test case	Preconditions	Test data	Steps	Expected result	Status
TP-11	NFR-01	Performance threshold	Test environment prepared; 50 concurrent authenticated sessions available.	Read endpoints defined in performance script.	Run workload and measure response times.	At least 95% of normal read requests complete within 2 s excluding external network latency.	Not Executed
TP-12	NFR-03	Allocation transaction integrity	Database transaction support enabled.	Room near capacity; allocation request.	Force a failure during allocation transaction.	Either the full allocation change commits or all related changes roll back; no partial state remains.	Not Executed
TP-13	NFR-04	Allocation workflow usability	Administrator is trained on the system.	Valid student and room.	Time/count the primary data-entry steps after login for a valid allocation.	Workflow completes in no more than five primary data-entry steps.	Not Executed
TP-14	SEC-01, SEC-03, SEC-04, SEC-05, SEC-06	Security control validation	Deployed test system with test accounts and logging.	Invalid credentials, unauthorized role, malformed inputs, HTTPS endpoint, allocation event.	Execute unauthorized-access, credential, transport, input-validation, and logging checks.	Controls are enforced; no plaintext credential is stored; insecure/unauthorized requests are rejected; required security events are logged.	Not Executed

5. Requirements Traceability Matrix

Requirement ID	Requirement area	Test / verification reference
FR-01	Authentication	TP-01, TP-14
FR-02	Student details management	TP-02
FR-03	Student search	TP-03
FR-04	Room details management	TP-04
FR-05	Valid allocation	TP-05
FR-06	Invalid allocation prevention	TP-06, TP-07
FR-07	Modify/deallocate	TP-08
FR-08	Occupancy	TP-09
FR-09	Student self-view	TP-10
NFR-01	Performance	TP-11

Requirement ID	Requirement area	Test / verification reference
NFR-02	System availability and recovery	NFR02_AVAILABILITY.md
NFR-03	Transaction integrity	TP-12
NFR-04	Usability	TP-13
NFR-05	Maintainability	Architecture inspection
SEC-01	Authentication control	TP-10, TP-14
SEC-02	Authorization	TP-10, TP-14
SEC-03	Password protection	TP-14
SEC-04	HTTPS/TLS	TP-14
SEC-05	Input/injection protection	TP-14
SEC-06	Security event logging	TP-14

6. Test Data Baseline

Data item	Example synthetic value	Purpose
Student identifiers	S001, S002	Positive, duplicate, and isolation cases.
Room identifiers	R101, R102	Allocation and occupancy cases.
Room capacity	1, 2, 3	Available and full-room conditions.
Roles	ADMIN, STUDENT	Authorization tests.
Allocation state	ACTIVE, DEALLOCATED	Lifecycle verification.

7. Execution Record

Execution evidence, actual results, defect identifiers, and final Pass/Fail status shall be added during test execution. No execution result is asserted by this document.

References

[1] ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*, IEEE, 2018.

[2] ISO/IEC/IEEE 29119-3:2021, *Software and systems engineering — Software testing — Part 3: Test documentation*, IEEE, 2021.

[3] ISO/IEC/IEEE 42010:2022, *Software, systems and enterprise — Architecture description*, IEEE, 2022.

[4] Software Engineering Mini-Project Deliverables Part-1, course handout, September 2026.